MILITARY SPECIFICATION

RESISTOR NETWORKS, FIXED, FILM, ESTABLISHED RELIABILITY
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-R-874, dated 29 June 1990, and is approved for use by all Departments and Agencies of the Department of Defense.

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3.2, delete and substitute the following:

"3.2 Qualification. Networks furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) at the time of award of contract (see 4.4 and 6.3). Authorized distributors that are approved to MIL-STD-790 distributor requirements by the QPL manufacturers are listed in the QPL."

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Add the following new paragraphs:

"3.5.2.1 Solder dip (retinning) leads. The manufacturer (or authorized category E or C distributor) may solder dip/retin the leads of product qualified to this specification only if the solder dip process has been approved by the qualifying activity.

"3.5.2.2 Qualifying activity approval. Approval of the solder dip process will be based on one of the following options:

a. When the original lead finish qualified was not solder dip lead finish 52 of MIL-STD-1276 (NOTE: The 200 microinch maximum thickness is not applicable). The manufacturer shall use the same solder dip process for retinning as is used in the original manufacture of the product.

b. When the lead originally qualified was not not solder dip lead finish 52 of MIL-STD-1276 as prescribed in 3.5.2.2(a), approval for the process to be used for solder dip shall be based on the following test procedure:

(1) Thirty samples of any resistance value for each style and lead finish are subjected to the manufacturer's solder dip process. Following the solder dip process, the resistors are subjected to the dc resistance test (and other group A electrical). No defects are allowed.

(2) Ten of the 30 samples are then subjected to the solderability test. No defects are allowed.

(3) The remaining 20 samples are subjected to the resistance to solder heat test, followed by the moisture resistance test. (NOTE: Solder dip of gold-plated leads is not allowed.)

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"3.5.2.3 Solder dip/retinning options. The manufacturer (or authorized category B or C distributor) may solder dip/retin as follows:

a. After the 100 percent group A screening tests. Following the solder dip/retinning process, the electrical measurements required in group A, subgroup 1, 100 percent screening tests shall be repeated on 100 percent of the lot. (NOTE: The manufacturer may solder dip/retin prior to the 100 percent electrical measurements of the group A subgroup 1 tests.) The percentage defective allowable (PDA) for the electrical measurements shall be as for the subgroup 1 tests.

b. As a corrective action if the lot fails the group A solderability test.

c. After the group A inspection has been completed. Following the solder dip/retinning process, the electrical measurements required in group A, subgroup 1, 100 percent screening test shall be repeated on 100 percent of the lot. The PDA for the electrical measurements shall be as for the subgroup 1 tests. Following these tests, the manufacturer shall submit the lot to the group A solderability test as specified in 4.8.6."

Add the following new paragraph:

"3.5.5 Bonding. If thermocompression bonding is used for internal connections, a gold to gold bond shall form a solid phase weld (see 3.1)."

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TABLE VII, delete and substitute:

TABLE VII. Failure rate substitution.

Failure rate	Failure rate substitution
S	---
R	S
P	R, P
M	S, R, P

TABLE VIII, delete and substitute:

TABLE VIII. Substitution of temperature characteristics.

Characteristic	Characteristic substitution
H	---
K	H
M	H, K

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TABLE IX, delete and substitute:

TABLE IX. Resistance tolerance substitution.

Resistance tolerance	Resistance tolerance substitution
B	---
D	B
F	B, D
G	B, D, F
J	B, D, F, G

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4.5.1, delete in its entirety.

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After 4.7.1.2.3 and before table XI, add the following new paragraphs:

"4.7.1.2.4 Subgroup 4 (solderability).

"4.7.1.2.4.1 Sampling plan. Thirteen samples shall be selected randomly from each inspection lot and subjected to the subgroup 4 solderability test. If there are one or more defects, the lot shall be considered to have failed.

"4.7.1.2.4.2 Rejected lots. In the event of one or more defects, the inspection lot is rejected. The manufacturer may use one of the following options to rework the lot:

a. Each production lot that was used to form the failed inspection lot shall be individually submitted to the solderability test as required in 4.8.6. Production lots that pass the solderability test are available for shipment. Production lots failing the solderability test can be reworked only if submitted to the solder dip procedure in b.

b. The manufacturer submits the failed lot to a 100-percent solder dip using an approved solder dip process per 3.5.2.1. Following the solder dip, the electrical measurements required in group A, subgroup 1 tests shall be repeated on 100 percent of the lot. The PDA for the electrical measurements shall be as for the subgroup 1 tests. Thirteen additional samples shall then be selected and subjected to the solderability test with zero defects allowed. If the lot fails this solderability test, the lot shall be considered rejected and shall not be furnished against the requirements of this specification.

"4.7.1.2.4.3 Disposition of samples. The solderability test is considered a destructive test and samples submitted to the solderability test shall not be supplied on the contract."

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TABLE XI, add the following new subgroup:

Inspection	Requirement paragraph	Test method paragraph	Sampling procedure
Subgroup 4			
Solderability	3.11	4.8.4	See 4.7.1.2.4

TABLE XII, delete and substitute the following table:

TABLE XII. Sampling plans for ppm categories.

Lot size	Sample size
2 to 8	100 percent
9 to 15	100 percent
16 to 25	100 percent
26 to 50	100 percent
51 to 90	100 percent
91 to 150	125
151 to 280	192
281 to 500	192
501 to 1,200	192
1,201 to 3,200	192
3,201 to 10,000	192
10,001 to 35,000	294
35,001 to 150,000	294
150,001 to 500,000	345
500,000 and over	435

TABLE XIII, delete:

4.7.1.3, delete and substitute:

4.7.1.3 Group B inspection. Group B inspection shall consist of the tests specified in table XIV, in the order shown. Tests shall be performed on sample units that have been subjected to and have passed the group A inspection.

TABLE XIV, delete solderability and related entries from subgroup 2.

30.3, delete and substitute:

30.3 Description of items. The manufacturer shall submit a detailed description of the network being submitted for inspection, including materials used for the resistance element and protective enclosure or coating. After qualification has been granted, no changes shall be made in materials, design, or construction without proper notification to the qualifying activity.

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Custodians:

40.1, delete and substitute:

40.1 Extension of qualification. The resistance range included in the qualification of any one network style will be between the lowest and highest resistance values which pass the qualification inspection (see 3.1). Qualification of one characteristic is basis for qualification of another characteristic, as indicated in table XIX. Qualification of the lower resistance tolerances will qualify for the higher resistance tolerances in accordance with table XX. Extent of qualification by style shall be in accordance with table XXI. Internal welded construction will qualify a nonwelded construction. Extent of qualification by schematic shall be in accordance with table XXII. As a requisite for extension of qualification as described herein between characteristics, tolerances, schematics, resistance values and failure rates, the product involved must be manufactured using the same facilities, processes, and materials as the product originally submitted for qualification.

TABLE XVIII, delete and substitute:

TABLE XVIII. Critical resistance value for qualification inspection. 1/

Style	Critical resistance value 1/ 2/					
	Schematic A	Schematic B	Schematic C	Schematic G	Schematic J	Schematic H
	Characteristics					
	H, K, M	H, K, M	H, K, M	H, K, M	H, K, M	H, K, M
	MΩ	MΩ	MΩ	MΩ	MΩ	MΩ
RZR10	0.047	0.1			.200	
RZR11	0.047	0.1			.200	
RZR20	0.047	0.1			.200	
RZR21	0.047	0.1			.200	
RZR30	0.047	0.1			.160	
RZR31	0.047	0.1			.160	
RZR40			.012	.012		.020
RZR41			.012	.012		.020
RZR50			.012	.012		.020
RZR51			.012	.012		.020
RZR60			.012	.012		.020
RZR61			.012	.012		.020
RZR70			.020	.020		.033
RZR71			.020	.020		.033
RZR80			.020	.020		.033
RZR81			.020	.020		.033
RZR90			.020	.020		.033
RZR91			.020	.020		.033
RZR95	0.047	0.10			.160	
RZR96	0.047	0.10			.160	

1/ Maximum continuous working voltage shall be applied (see 3.1).
2/ The critical resistance value is the maximum standard resistance value which will dissipate full wattage when maximum continuous working voltage is applied.

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TABLE XXI, delete and substitute:

TABLE XXI. Extent of qualification by style.

Style	Will qualify style(s)
RZR10	RZR10
RZR11	RZR11, RZR10
RZR20	RZR20, RZR10
RZR21	RZR21, RZR20, RZR11, RZR10
RZR30	RZR30
RZR31	RZR31, RZR30
RZR40	RZR40
RZR41	RZR41, RZR40
RZR50	RZR40, RZR50
RZR51	RZR51, RZR50, RZR41, RZR40
RZR60	RZR60, RZR50, RZR40
RZR61	RZR61, RZR60, RZR51, RZR50, RZR41, RZR40
RZR70	RZR70
RZR71	RZR71, RZR70
RZR80	RZR80, RZR70
RZR81	RZR81, RZR80, RZR71, RZR70
RZR90	RZR90, RZR80, RZR70
RZR91	RZR91, RZR90, RZR81, RZR80, RZR71, RZR70
RZR95	RZR30
RZR96	RZR96, RZR95, RZR31, RZR30

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity
Army - ER

Agent:
DLA - ES

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